

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop Dynamic Webpage Using Java Script

2. AIM

To develop a **dynamic webpage using HTML5, CSS3, and JavaScript** by using **DOM manipulation, JavaScript functions, and event handling** to change text, background color, and display the current date and time.

3. OBJECTIVE

- To understand the basics of JavaScript.
 - To use JavaScript functions and events.
 - To modify HTML content dynamically.
 - To change webpage styles using JavaScript.
 - To display the current date and time.
 - To understand DOM manipulation ○ .
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4. THEORY

JavaScript is a scripting language used to make webpages interactive and dynamic. It can modify HTML elements and CSS styles based on user actions.

In this program, JavaScript uses **DOM (Document Object Model)** methods to dynamically change the webpage.

- `getElementById()` is used to access an HTML element.
- `innerHTML` is used to change the content of an element.
- `style.backgroundColor` is used to change the background color.
- `Date()` is used to obtain the current date and time.
- `onclick` executes a JavaScript function when a button is clicked.

The basic flow is:

User Action → JavaScript Function → DOM Manipulation → Updated Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML file.
 2. Add the basic HTML5 structure.
 3. Create a heading and three buttons.
 4. Create a paragraph to display the current time.
 5. Add CSS to style the webpage and buttons.
 6. Create the `changeText()` function to change the heading.
 7. Create the `changeColor()` function to change the background color.
 8. Create the `showTime()` function to display the current date and time.
 9. Use `onclick` events to call the functions.
 10. Save and open the file in a web browser.
 11. Click each button and verify the output.
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6. PROGRAM

Html&css.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Webpage Using JavaScript</title>
<style>    body{  font-family: Arial, sans-serif;
background-color: rgb(173, 230, 206);  text-align:
left;  margin: 0;
}
header{
backgroundcolor: navy;
color: white;  padding:
20px;
}
.container{  margintop:
50px;
}
button{  padding: 10px
20px;  margin: 10px;
background-color: green;
color: white;  border:
none;  border-radius: 5px;
cursor: pointer;
}
button:hover{  background-color:
rgb(236, 10, 44);
} p{  font-size:
20px;  color:
white;
margintop: 20px;
}
</style>
</head>
<body>
  <header>
    <h1>Dynamic Webpage Using JavaScript</h1>
  </header>

```

```

<div class="container">
  <h2 id="message">Welcome to JavaScript</h2>
  <button onclick="changeText()">Change Text</button>
  <button onclick="changeColor()">Change Background</button>
  <button onclick="showTime()">Show Time</button>
  <p id="time"></p>
</div>
<script>
  // Change the heading text function changeText() {
document.getElementById("message").innerHTML =
  "JavaScript Makes Webpages Dynamic!";
}

// Change the background color function changeColor()
{  document.body.style.backgroundColor = "orange";
}

// Display current date and time function showTime() {  let now = new
Date();  document.getElementById("time").innerHTML =
"Current Time: " + now.toLocaleString();
}
</script>

</body>
</html>

```

7. CODE EXPLANATION

Element / Code	Description
<button onclick="changeText()">	Calls the changeText() function
getElementById("message")	Selects the heading element.

innerHTML	Changes the heading text.
document.body.style.backgroundColor	Changes the webpage background color.
new Date()	Gets the current date and time.
toLocaleString()	Formats the date and time.
getElementById("time")	Selects the paragraph where time is displayed.
onclick	Performs an action when the button is clicked.
:hover	Changes the button appearance when the mouse is over it.

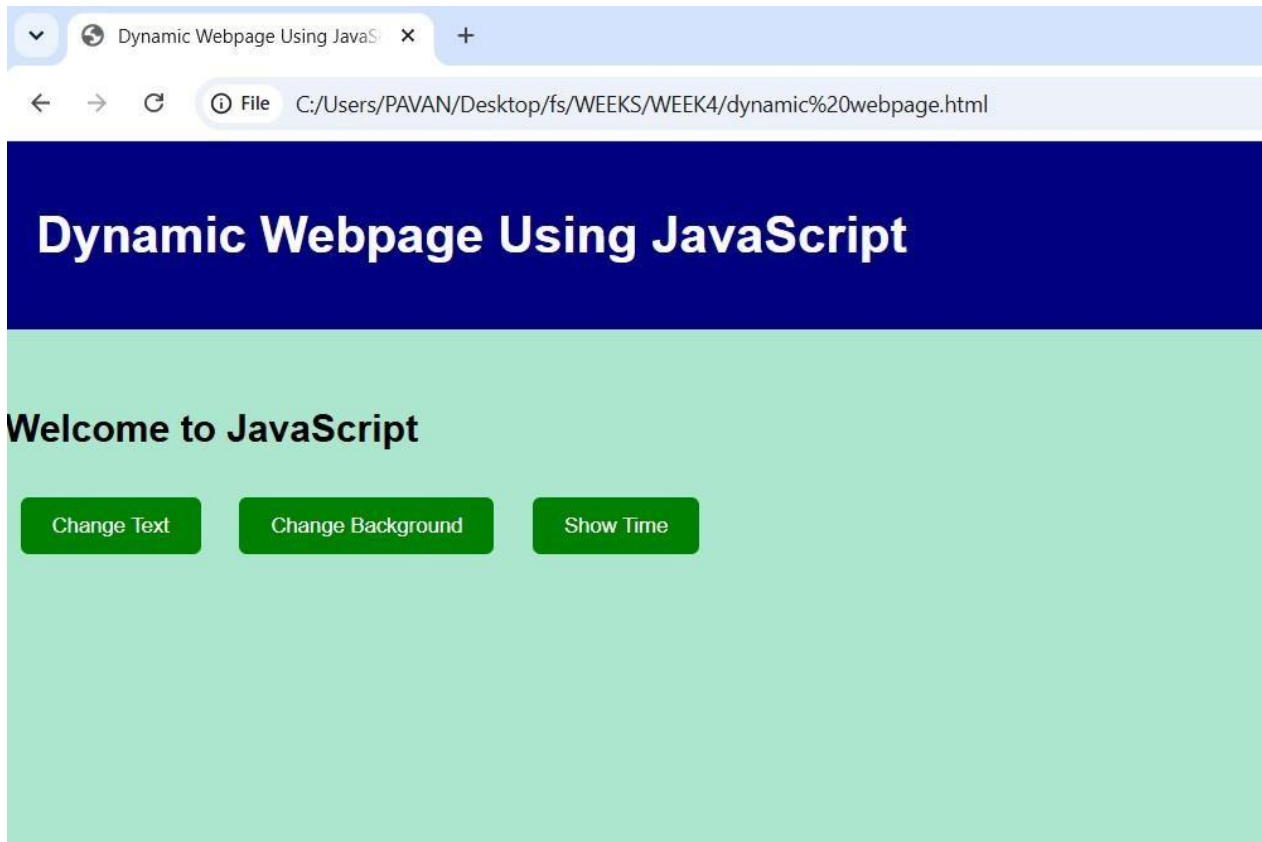
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the webpage	Webpage should display correctly	Webpage displayed	Pass
TC-02	Click Change Text	Heading text should change	Heading text changed	Pass
TC-03	Click Change Background	Background color should change to orange	Background changed to orange	Pass
TC-04	Click Show Time	Current date and time should be displayed	Date and time displayed	Pass

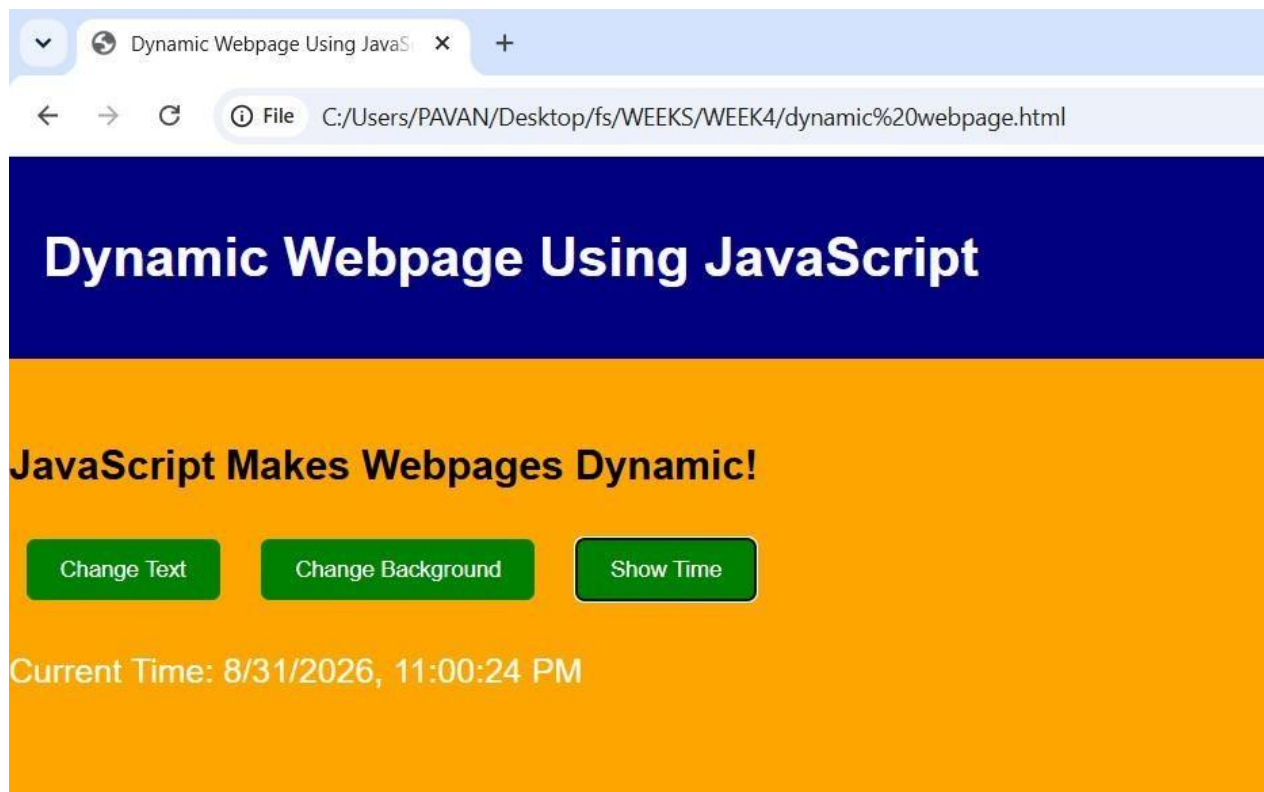
9. OUTPUT

Output :

Before clicking the buttons



After clicking the Buttons



10. RESULT

Thus, a **dynamic webpage using JavaScript** was successfully developed. The webpage dynamically changes the heading text, background color, and displays the current date and time when the corresponding buttons are clicked..
